

Palleti Vamshi

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EDUCATION

Vallurupalli Nageswara Rao Vignana Jyothi Institute of Engineering and Technology (VNR VJIE)

Bachelor of Technology (B.Tech) in Artificial Intelligence & Machine Learning (Current: 2nd Year)

Hyderabad, India

Graduation: 2029

Cumulative GPA: 9.9 / 10

TECHNICAL SKILLS

Programming: C, C++, Python, Java (21 LTS), JavaScript (ES6+)

AI & Machine Learning: Scikit-Learn, XGBoost, Machine Learning Foundations, Explainable AI Concepts

Backend Development: Node.js, Express.js (5.x), Spring Boot (4.x), Spring Security (7.x)

Databases & Persistence: MongoDB, MySQL (3NF Relational Schema Design), H2 In-Memory

Tools, Systems & Protocols: Git, GitHub, Apache Maven, Vite, MQTT (Mosquitto & Paho), Linux CLI

TECHNICAL PROJECTS

LightX-IDS (Evolution of IDS prototype) [[GitHub !+](#)] *Python, MQTT (Mosquitto), Scikit-Learn, XGBoost, Digital Twin Simulation*

- Constructed an Industrial Digital Twin simulating 6 machine classes (Motor, Pump, Tank, Conveyor, Valve, Compressor) with 10 continuous telemetry sensor types.
- Engineered an event-driven telemetry pipeline streaming sensor packets over MQTT (Mosquitto) to ingest and preprocess structured industrial datasets.
- Developed model training routines for lightweight tree-based classifiers (Decision Trees, Random Forest, XGBoost) targeting low-latency edge deployment and Explainable AI (SHAP).

Smart Campus Management System [[GitHub !+](#)] *Java 21, Spring Boot 4.1, Spring Data JPA, MySQL 8.x, H2 Database, Maven*

- Designed an enterprise-grade 16-table normalized relational MySQL schema (3NF) modeling departments, courses, faculty, students, daily attendance, and certificate requests.
- Implemented 16 Spring Data JPA repositories with Jakarta Validation and Hibernate schema validation (ddl-auto=validate), enforcing composite unique constraints and referential integrity.
- Created complete database seed datasets and 13 complex DBMS queries verifying timetable conflict detection, validated against automated in-memory H2 tests.

Mentor-Student Management [[GitHub !+](#)] *Node.js, Express 5.2, MongoDB, Mongoose 9.1, JWT, BCrypt*

- Developed a modular asynchronous REST backend service using Express 5 and ES modules for academic mentorship tracking during the P17 Hackathon.
- Configured MongoDB persistence using Mongoose 9 schemas, enforcing strict schema validation and document lifecycle management across mentor and student allocations.
- Implemented stateless token authentication using JSON Web Tokens (JWT), BCrypt credential hashing, and HTTP cookie parsing middleware.

Attendance Management System | *Python, File I/O, Structured Data Processing, CSV Reporting*

- Built a standalone Python application to automate student attendance capture, roster verification, and record management.
- Developed validation logic to eliminate duplicate roll number submissions and compute session-wise as well as cumulative attendance statistics.
- Implemented file persistence routines and formatted report exports to streamline administrative record-keeping.

LEADERSHIP & EXTRACURRICULAR ACTIVITIES

AWS Student Builder Group — Active Member: Engaged in collaborative technical learning sessions, cloud architecture discussions, and peer programming workshops.

Department Academic & Technical Events — Technical Operations Contributor: Contributed to the coordination of technical hackathons, resource scheduling, and student event logistics.

CORE STRENGTHS

Problem Solving • Critical Thinking • Consistency • Leadership • Collaboration • Technical Curiosity